COMPRITOL® 888 ATO LIPID MATRICES FOR SUSTAINED RELEASE PROCESSED BY HOT-MELT EXTRUSION

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ABSTRACT SUMMARY

In the current study we explored the efficiency of Compritol® 888 ATO, a lipid excipient, for the manufacture of sodium diclofenac (DfNa) sustained release matrices by using Hot Melt Extrusion (HME) processing. The extrudates were processed with a twin screw Eurolab 16 extruder and the pellets obtained were further milled to produce granules.

Different HME processing approaches were used including "cold" extrusion where drug/lipid binary blends at different ratios were processed at temperatures below the lipid's melting point. The blends were also extruded at temperatures above the lipids' melting point and the extruded granules were used in tablet formulations made by direct compression at different compaction forces. In addition pre-blended tablet formulations were extruded to investigate the effect on drug release. Furthermore, the extrudates were characterized by X-ray powder diffraction (XRPD), differential scanning calorimetry (DSC) and hot stage microscopy (HSM) in order to identify the state of the active substance within the lipid matrix.

INTRODUCTION

HME extrusion has been used for the development of various solid dosage forms aiming to increase the solubility of poorly soluble drugs, taste mask bitter substances and control drug release mainly by using thermoplastic polymers after the addition of plasticizers.

There are only a few studies refering to lipid extrusion for taste masking and sustained release where, during the HME processing the materials were extruded mainly at temperatures below the melting point of the lipid[1,2].

Glyceryl behenate (Compritol® 888 ATO) is a hydrophobic fatty acid ester of glycerol with a HLB of 1 and a melting point 70°C. It is a fine white powder with a mean particle diameter of 50 µm used in tablet processing as a lubricant or as a lipid matrix former for sustained release. The mechanism of drug release is primarily diffusion through the insoluble, non-swellable matrix. In the current study, the effect of the drug/lipid ratio and the processing parameters on the dissolution rate of DfNa was investigated.

EXPERIMENTAL METHODS

The DfNa/Compritol® 888 ATO blends (at ratios of 30:70, 40:60 and 50:50, Table 1) were mixed in 100 g batches for 10 min in a Turbula TF2 mixer. The extrusion was performed using a Eurolab 16 twin screw extruder using various temperatures and 50 rpm screw speed. The extrudates were milled by ball milling with a rotational speed of 200 rpm for 2 mins each.

The extruded granules were then blended with standard tablet excipients and compressed under different compaction forces using a Flexitab tablet press to produce tablets with 10, 15 and 20kP hardness strength. The granules were characterized by XRPD, DSC, HSM and dissolution studies were carried out in 900 ml distilled water at 37°C, 100 rpm.

Table 1: Composition of tablet formulations process by HME.

Materials	Batch A 30:70 (%)	Batch B 40:60 (%)	Batch C 50:50 (%)
DfNa/Compritol® 888 ATO extr.	55	55	55
Dibasic calcium phosphate anhydrous (Fujicalin)	14.0	14.0	14.0
Lactose (Lactopress)	27.0	27.0	27.0
Magnesium alumino metasilicate (Neusilin)	3.0	3.0	3.0
Magnesium stearate	1.0	1.0	1.0

RESULTS AND DISCUSSION

Due to the properties of Compritol® solid extrudates (Batches A, B and C) of DfNa were initially prepared by "cold" extrusion processing at temperatures below the melting point of Compritol® (<58°C).
The same batches were processed at extrusion temperatures above the melting point of Compritol® (73°C).
The pre-mixed formulations were extruded at higher temperatures because of the presence of other excipients (e.g. fujicalin, neusilin) with maximum processing temperatures of 85°C.

Fig. 1 shows the X-ray patterns of the extruded granules; the DfNa intensity peaks indicate the presence of crystalline drug within the polymer matrix.
Similar intensity peaks were observed for higher DfNa ratios even after the processing of the pre-mixed batches. As a result DfNa is not solubilised in the lipid matrix during the HME processing.

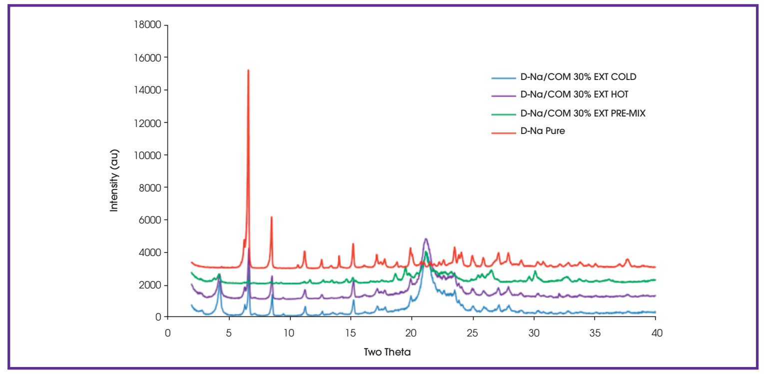

Fig. 1: X-ray diffraction patterns of pure DfNa and extruded DfNa/Compritol® 888 ATO (30:70) granules.

Fig. 3 shows the effect of the HME processing on the DfNa dissolution profiles. Drug dissolution varied depending on the extrusion process, the drug/lipid ratio and the tablet hardness.

The "cold" extruded tablets showed faster dissolution rates which became slower at higher compaction forces. The increase of the drug content from 30-50% was associated with faster dissolution rates. Similar results were observed for each HME. However, the extruded pre-mixed batches showed slower and more stable dissolution rates. In addition the texture of the compressed tablets was substantially improved showing a gloss finish.

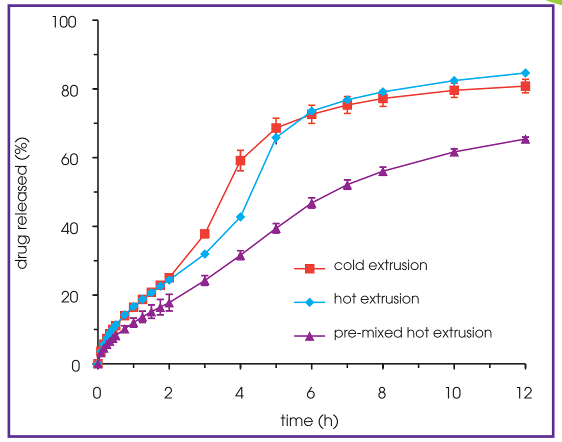

Fig. 3: Dissolution profiles of DfNa/Compritol® 888 ATO (40:60) from lipid matrix tablets (20kN hardness).

Figure 4 illustrates the tablet's aspect after 12h of dissolution. It can be seen that tablets prepared by the hot melt extrusion process (left hand side) disintegrated upon contact with the release medium, which explains increased drug release after 4h (Figure 3).

Extruding the whole tablet formulation (pre-mix + hot extrusion, right hand side) led to improved tablet robustness and thus sustained drug release over at least 12h.

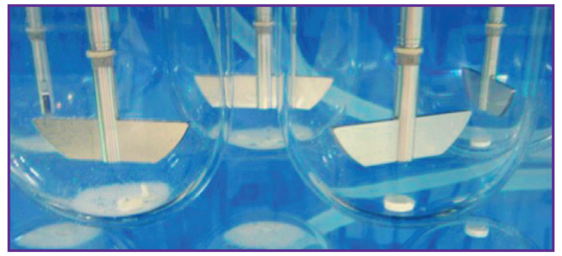

Fig. 4: Photo of dissolution of tablet made by hot extrusion process (left hand side vessel); tablet made by hot extruded + pre-mix process (right hand side vessel) after 12 h dissolution.

The XRPD results were confirmed by DSC scans of the extruded batches. In Fig. 2b, DfNa melting endothermic peaks were shifted to lower temperatures (238-243°C) compared to the pure substance. Even at high DfNa loading ratios (50/50) a clear endothermic peak can be observed.

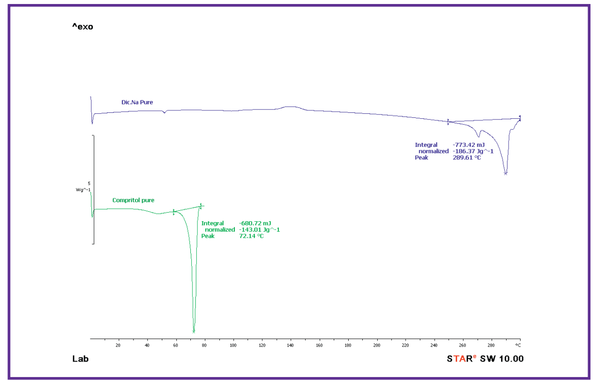

Fig. 2a: DSC scans of pure DfNa, Compritol® 888 ATO.

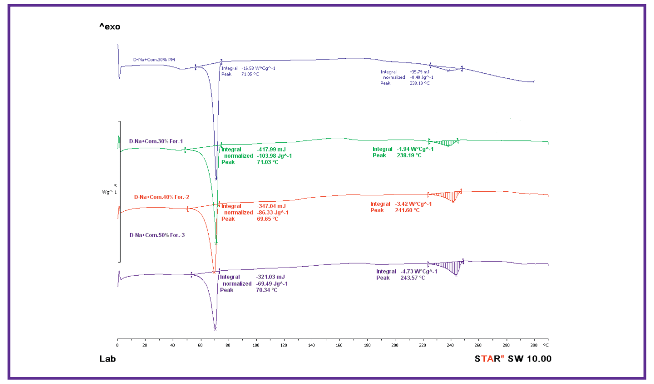

Fig. 2b: DSC scans of hot extruded DfNa/Compritol® 888 ATO granules at different ratios.

CONCLUSION

The current study demonstrated the efficiency of HME to produce solid dispersions of drug/lipid excipient for sustained release at various extrusion conditions at high drug loading. The processing conditions can have an effect on the drug release profile and these can be adjusted along with the tablet properties to provide the desired release patterns.

HME processing with Compritol® 888 ATO can be effectively used to provide sustained drug release of water soluble actives such as DfNa.

REFERENCES

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